AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 20/Nov./2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1
(10 marks - 10 MCQ questions)

1- The Advanced Research Projects Agency Network (ARPANet) project of 1969 was aimed at creating _____ that would be resistant to failure.

A. fiber optics cables.
B. computer networks.
C. satellites.
D. protocols.

2- W3C sees the development of the _____ initiative as, a solution to providing internet for remote communities. This means creating a pared down version of XHTML that can be readily interpreted by many different devices. To this end the W3C created the XHTML Basic document type.

A. Mobile Web.
B. XHTML Basic.
C. Metadata.
D. SGML.

3- The goal in designing the _____ of any website is to ensure that visitors can find what they want quickly and efficiently so that they will come back for more.

A. look and find.
B. Browse and look.
C. log and look.
D. look and feel.

4- The _____ attribute provides a mechanism for attaching a short text equivalent to any visual or audible embedded object within a web page.

A. longdesc.
B. attach.
C. add.
D. alt.

5- Typography is the _____.

A. art of selecting color palettes and layout grids to enhance visual storytelling across digital platforms.
B. practice of arranging images and illustrations to create emotional impact and guide viewer attention.
C. balance and interplay of letterforms on the page – a verbal and visual equation that helps the reader understand the form and absorb the substance of the page content.
D. system of encoding linguistic meaning through symbols – often used in branding and logo design to convey identity.

6- When dealing with forms, the name attribute is included to _____.

A. locate the script that will process the form data and apply the formatting of the CSS file.
B. identify the form to the script at the server and also can be used to identify the form element for CSS.
C. pass the encrypted data in the http response stream to the server.
D. accept the type of defined name in the head section of the web page.

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7- The architecture of web applications in terms of their structure, _____
A. which can relate to hardware (the physical structure of the application), software (the programs making up an application), and other components such as clouds computing, blueprints, guidelines, leadership, etc.
B. which can relate to a term normally associated with the design of buildings. In the construction industry, architecture is often taken to relate to the style and structure of buildings.
C. which can relate to both hardware (the physical structure of the application) and software (the programs making up an application).
D. which can relate to the protocols that are supporting the passing of messages between components, provides a standard approach to encoding information in messages, describing a web service so that it can be accessed and used, and publish descriptions of and find web services in a central repository.

8- "Plugs-ins" is _____.
A. a type of hardware device that connects to a computer to enhance its physical capabilities.
B. a software extension that adds specific functionality to a browser or application.
C. a built-in feature of operating systems used to manage user accounts and permissions.
D. a programming language used to create interactive web pages and dynamic content.

9- The advantage of using Objects in JavaScript regular expression is _____.
A. that the expression value can be created at runtime whereas literals, more efficiently, are compiled when the page is loaded.
B. that objects automatically escape special characters, making expressions safer for user input.
C. that regular expression objects are faster than literals because they are precompiled during script execution.
D. that using object allows regular expressions to be embedded directly into HTML without additional syntax.

10- In the following code:

```
if (!$connection) {
    Print ("Internal error". mysql_errno()); }:
```

A. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case an error message is printed and the 'mysql_error()' will be called.
B. The code has a conditional expression '!' (i.e. not) connection that will be false if 'connection' evaluates to true. In this case an error message is printed and the 'mysql_error()' will be called.
C. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case, the connection will be established.
D. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case, the connection will be established.

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PART 2
(20 marks - 6 short questions)

Question 1: (3 marks)
Discuss the types of web applications in terms of their web content.

Question 2: (3 marks)
What is the difference between Elements and Attributes in HTML? show an example for each one.

Question 3: (3 marks)
Discuss three quality components that define Usability.

Question 4: (3 marks)
According to the W3C, accessibility resources are divided into three main areas. List these areas.

Question 5: (4 marks)
There are a number of simple design principles you should follow when designing for the web, list these principles.

Question 6: (4 marks)
List Service-oriented architecture properties.

PART 3
(30 marks - 5 Problem Solving Questions

Question 1: (5 marks)
Assume that a user is visiting a website with an average data transmission rate of 64 (Kbps); consider that the request has the following:

1) Three webpages (HTML tags) at an average size of 2500 bytes for each file.
2) Two CSS files at an average size of 4000 bytes for each file.
3) Four images at an average size of 1500 bytes for each file.
4) One banner image at a size of 1000 bytes.

Assume that it takes the server 2 milliseconds to respond to each file request.
Calculate the total time required to download all these files.
Note:
1 Kbps = 1000 bits/second,
1 byte = 8 bits
1 second = 1000 millisecond.

Question 2: (5 marks)
Write the HTML code to create an ordered list that includes the following items:

1. Bahrain
2. Egypt
3. Lebanon
4. Oman

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Question 3: (5 marks)

What is the output of the following code?

```
<html>  <script>
 let x = 3;
 let output = "";
 for (let y = 2; y <= 6; y++) {
  if ((x + y) % 3 == 0) {
    x += 4;
  } else if (x % 2 != 0 && y % 2 == 0) {
    x -= 1;  }
    else {
    x *= 2;  }
  output += "x=" + x + ", y=" + y + "<br>";   }
 document.write(output);
</script>  </html>
```

Question 4: (5 marks)

Write a JavaScript program that performs the following tasks:

1) Prompts the user to enter the age of 5 people.
2) Classifies each person into one of the following categories:
   2.1. Child (age < 13)
   2.2. Teenager (13 ≤ age < 20)
   2.3. Adult (20 ≤ age < 60)
   2.4. Senior (age ≥ 60)
3) Prints a summary showing how many people fall into each category.

Question 5: (10 Marks)

Develop a simple web page containing the following specification:

1. Page Title "TT284 Group Leader Form"
2. Header 1 "TT284 Group Leader Details"
3. Form action to locate the script that will process the form data "http//norton.open.ac.uk/reflect.php"
4. Text Input type field for the Leader Name.
5. Checkbox type to select the preferred Training Type as follows:
   a) Practical Training.
   b) Theoretical Training.
6. List type to select Venue as follows:
   a) Labs.
   b) Online.
7. Submit button.
8. Reset button.

The web page should look like the following design (Figure 1):

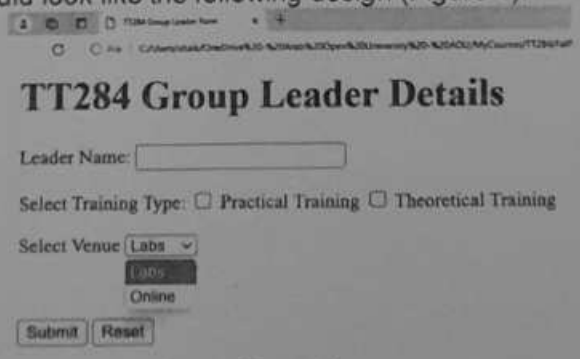


Figure 1

End of questions

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